413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 15

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Recall

In Lecture 14, we saw that increasing model complexity can only decrease the training residual sum of squares,

$$\mathrm{RSS}_{p+1} \leq \mathrm{RSS}_p,$$

but a smaller training error does not necessarily imply better prediction on new data. More flexible models can have lower bias but higher variance. We also introduced regularization and cross-validation as two ways to control model complexity:

$$\hat{\beta}_\lambda = \arg\min_\beta \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\},$$

with the complexity parameter chosen using cross-validation rather than training loss/error alone. This leads to the central question for this lecture: **Can model complexity be assessed directly from the fitted model, without a separate validation split?**

1 Adjusted R^2: Accounting for Model Complexity

In Lecture 13 and 14, we saw that increasing model complexity can only improve, or leave unchanged, the fit to the training data. For nested regression models,

$$\mathrm{RSS}_{p+1} \leq \mathrm{RSS}_p.$$

Recall that

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}.$$

Therefore, adding more predictors can never decrease the training R^2. This creates an immediate problem for model selection. If we compare models only using R^2, the most flexible candidate model will always appear at least as good as every simpler model, even when the additional predictors explain only sample-specific noise.

1.1 Adjusted R^2

Suppose the regression model contains p predictors, not counting the intercept, and is fitted to n observations. The adjusted coefficient of determination is defined as

$$R^2_{\mathrm{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1}.$$

Using the relation $1 - R^2 = \frac{\mathrm{RSS}}{\mathrm{TSS}}$, we can write

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)}.$$

This form reveals the motivation for adjusted R^2. The quantity

$$\frac{\mathrm{RSS}}{n-p-1}$$

is the residual mean square. When an additional predictor is introduced, two things happen:

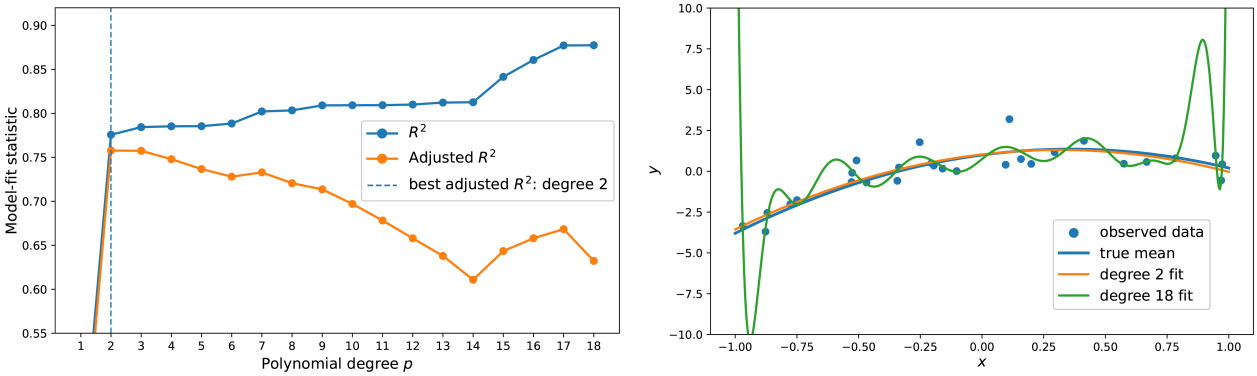


Figure 1: Comparison of ordinary R^2 and adjusted R^2 as polynomial model complexity increases. The ordinary R^2 increases monotonically with polynomial degree because each higher-degree model contains the lower-degree model as a special case and therefore cannot increase the training RSS. In contrast, adjusted R^2 accounts for the loss of residual degrees of freedom and reaches its maximum at degree $p = 2$. The right panel illustrates why: although the degree-18 polynomial achieves a better training fit, it exhibits strong oscillations and fits sample-specific fluctuations, whereas the degree-2 model better reflects the underlying conditional mean.

$$\mathrm{RSS} \downarrow, \qquad n - p - 1 \downarrow .$$

The new predictor therefore improves adjusted R^2 only if the reduction in RSS is sufficiently large to compensate for the loss of one residual degree of freedom. Hence, unlike ordinary R^2, R^2_{adj} **can decrease when an unnecessary predictor is added.**

1.2 Example: Polynomial Degree

Let us return to polynomial regression. Given observations (x_i, y_i), we seek to approximate the relationship between x and y using a polynomial of unknown degree. We therefore consider the nested sequence of models

$$\mu_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p, \qquad p = 1, 2, \ldots, 18.$$

Because the degree-$(p+1)$ model contains the degree-p model as a special case,

$$\mathrm{RSS}_{p+1} \leq \mathrm{RSS}_p.$$

Therefore,

$$R^2_{p+1} \geq R^2_p.$$

Hence, ordinary R^2 cannot determine an appropriate polynomial degree: maximizing R^2 always selects the largest candidate model. Adjusted R^2 introduces a complexity correction. An additional polynomial term increases adjusted R^2 only if the resulting reduction in RSS is large enough to compensate for the loss of one residual degree of freedom.

1.3 From Adjusted R^2 to Information Criteria

Adjusted R^2 introduces an important principle for model selection:

improved fit should be balanced against increased model complexity.

Unlike ordinary R^2, adjusted R^2 does not automatically reward the inclusion of additional predictors. A new predictor improves adjusted R^2 only if the resulting reduction in residual error is large enough to compensate for the loss of a degree of freedom. This idea is useful, but adjusted R^2 is tied specifically to least-squares regression and to the residual sum of squares. We would like a more general framework that can be applied to other probabilistic models as well.

A natural way to do this is to replace the residual sum of squares by the likelihood. For a fitted statistical model, the maximized likelihood measures how well the model explains the observed data, while the number

of free parameters measures its complexity. This leads to information criteria such as AIC and BIC, which retain the same basic principle:

$$\textbf{goodness of fit} \quad \text{versus} \quad \textbf{model complexity}.$$

The main difference is that AIC and BIC express goodness of fit through the maximized likelihood and impose an explicit penalty for the number of estimated parameters.

1.4 Akaike Information Criterion (AIC)

The Akaike Information Criterion is defined as

$$\text{AIC} = -2 \log \hat{L} + 2k,$$

where $\hat{L}$ is the maximized likelihood of the fitted model and k is the number of estimated parameters. The criterion consists of two competing terms:

- $-2 \log \hat{L}$ measures the lack of fit. A smaller value indicates better agreement with the observed data.
- $2k$ penalizes model complexity. Each additional parameter increases the penalty by 2.

AIC balances the maximized likelihood against a complexity penalty $2k$; the preferred model is the one with the smallest AIC.

1.5 Bayesian Information Criterion (BIC)

The Bayesian Information Criterion takes a similar form:

$$\text{BIC} = -2 \log \hat{L} + k \log n,$$

where n is the number of observations. The main difference is the complexity penalty. Whereas AIC penalizes each parameter by 2, BIC penalizes each parameter by $\log n$. For example, when $n = 100$,

$$\log n \approx 4.61.$$

Thus, BIC imposes a stronger penalty for additional parameters than AIC in this example. For $n > e^2$, BIC imposes a larger per-parameter penalty than AIC and therefore tends to favor simpler models.

AIC and BIC are defined in terms of the maximized likelihood and therefore apply beyond linear regression. For Gaussian regression, however, the maximized likelihood can be written explicitly in terms of the residual sum of squares, which connects these criteria directly to the regression quantities introduced earlier.

1.6 Likelihood for Gaussian Regression

Consider the Gaussian linear regression model

$$Y_i = x_i^T \beta + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

or equivalently,

$$Y_i \mid x_i \sim \mathcal{N}(x_i^T \beta, \sigma^2).$$

For independent observations

$$(y_1, x_1), \ldots, (y_n, x_n),$$

the likelihood is

$$L(\beta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - x_i^T \beta)^2}{2\sigma^2} \right].$$

Taking log and using the definition of $\text{RSS}(\beta) = \sum_{i=1}^{n} (y_i - x_i^T \beta)^2$, we obtain

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2.$$

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\mathrm{RSS}(\beta).$$

For fixed σ^2, the first two terms are independent of β. Hence,

$$\arg\max_{\beta} \ell(\beta, \sigma^2) = \arg\min_{\beta} \mathrm{RSS}(\beta),$$

so the maximum-likelihood estimate of β coincides with the ordinary least-squares estimate:

$$\hat{\beta}_{\mathrm{MLE}} = \hat{\beta}_{\mathrm{OLS}}.$$

The variance parameter σ^2 must also be estimated. Differentiating the log-likelihood with respect to σ^2 gives

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{RSS}(\beta)}{2(\sigma^2)^2}.$$

Setting the derivative equal to zero,

$$-\frac{n}{2\sigma^2} + \frac{\mathrm{RSS}(\beta)}{2(\sigma^2)^2} = 0,$$

which implies

$$\hat{\sigma}^2(\beta) = \frac{\mathrm{RSS}(\beta)}{n}.$$

Evaluating this expression at $\hat{\beta}_{\mathrm{MLE}}$ gives

$$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{n}.$$

This differs from the unbiased estimator of the noise variance, which divides the RSS by the residual degrees of freedom. For likelihood-based model comparison, however, the relevant quantity is the maximum-likelihood estimate above. We now evaluate the log-likelihood at

$$\hat{\beta}_{\mathrm{MLE}} \qquad \text{and} \qquad \hat{\sigma}^2_{\mathrm{MLE}}.$$

Let

$$\hat{\ell} = \ell(\hat{\beta}_{\mathrm{MLE}}, \hat{\sigma}^2_{\mathrm{MLE}}).$$

Then

$$-2\hat{\ell} = n\log(2\pi) + n\log(\hat{\sigma}^2_{\mathrm{MLE}}) + \frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{\hat{\sigma}^2_{\mathrm{MLE}}}.$$

Since

$$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{n},$$

the last term satisfies

$$\frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{\hat{\sigma}^2_{\mathrm{MLE}}} = n.$$

Therefore,

$$-2\hat{\ell} = n\log\left(\frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{n}\right) + n\log(2\pi) + n.$$

For models fitted to the same observations, the terms

$$n\log(2\pi) + n$$

are identical across models. Thus, for model comparison,

$$\boxed{-2\hat{\ell} \equiv n\log\left(\frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{n}\right),}$$

where $\equiv$ denotes equality up to a constant common to all candidate models. Hence, in Gaussian regression, the maximized log-likelihood can be expressed directly in terms of the residual sum of squares. This provides the likelihood-based measure of fit used in AIC and BIC.

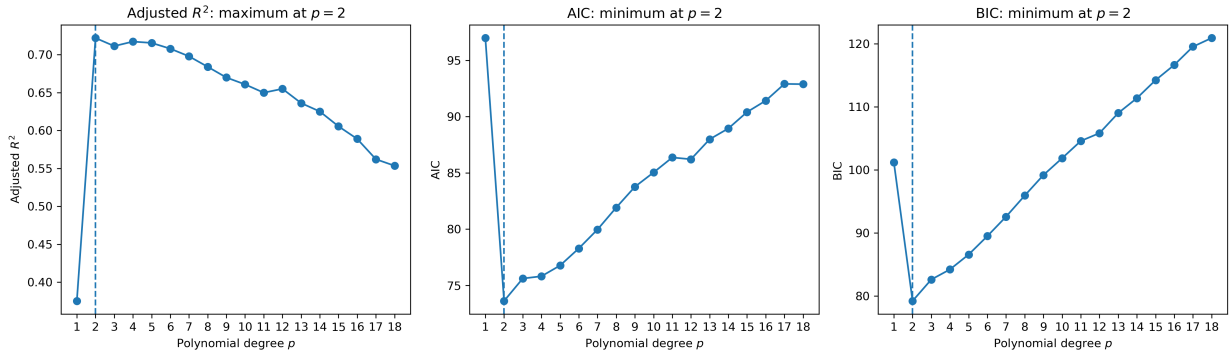


Figure 2: Model selection for polynomial regression using adjusted R^2, AIC, and BIC. Synthetic data were generated from the quadratic conditional mean $\mu(x) = 1+2x-2.8x^2$, with $n = 30$ observations and Gaussian noise with standard deviation $\sigma = 0.7$. Polynomial models of degree $p = 1, \ldots, 18$ were fitted to the same data. Adjusted R^2 is maximized at $p = 2$, while both AIC and BIC are minimized at $p = 2$. Thus, despite the continued reduction in training RSS for higher-degree models, all three criteria select the quadratic model.

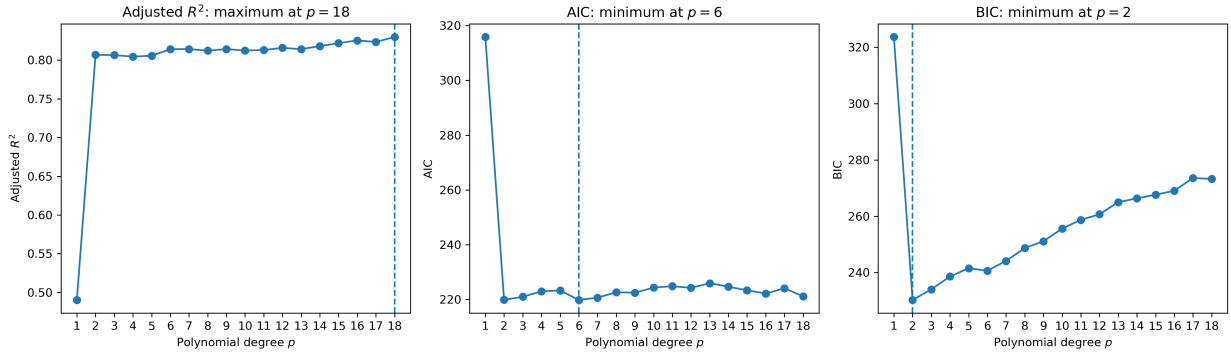


Figure 3: Model selection for polynomial regression with $n = 100$ observations, using the same quadratic conditional mean and noise level as in the previous example. Polynomial models of degree $p = 1, \ldots, 18$ were fitted to the data. Adjusted R^2 is maximized at $p = 18$, AIC is minimized at $p = 6$, and BIC is minimized at $p = 2$.

1.7 AIC and BIC in Terms of the Residual Sum of Squares

For Gaussian regression, we have therefore reduced the likelihood-based measure of fit, $-2\hat{\ell}$, to a function of the residual sum of squares. Substituting this expression into the definitions of AIC and BIC gives their corresponding forms for Gaussian linear regression.

$$\text{AIC} \equiv n \log\left(\frac{\text{RSS}}{n}\right) + 2k$$

and

$$\text{BIC} \equiv n \log\left(\frac{\text{RSS}}{n}\right) + k \log n.$$

Here $k = p + 2$ for a model with p predictors, an intercept, and an estimated error variance. Both criteria illustrate the same fundamental trade-off:

$$\underbrace{n \log\left(\frac{\text{RSS}}{n}\right)}_{\text{Goodness of fit}} + \underbrace{\text{Complexity penalty}}_{\text{Prevent overfitting}}.$$

Increasing model complexity generally reduces the RSS, improving the first term. However, it also increases the complexity penalty. The preferred model is therefore not necessarily the model with the smallest training RSS, but the one that achieves the most favorable balance between goodness of fit and complexity.

1.8 Comparison of Model Selection Criteria

Figures 2 and 3 illustrate that adjusted R^2, AIC, and BIC can select different model complexities. For $n = 30$, all three criteria select the quadratic model, $p = 2$, as shown in Figure 2. In this case, the improvement in fit obtained by adding higher-order polynomial terms is not sufficient to overcome the corresponding complexity penalties.

For $n = 100$, the criteria no longer agree. Figure 3 shows that adjusted R^2 is maximized at $p = 18$, AIC is minimized at $p = 6$, and BIC is minimized at $p = 2$. The different selections made by AIC and BIC reflect their different complexity penalties.

$$2k \qquad \text{versus} \qquad k \log n.$$

For $n = 100$,

$$\log n \approx 4.61 > 2,$$

so BIC imposes a larger penalty for each additional parameter and therefore favors a simpler model than AIC in this example. Thus, the selected model depends on the criterion used to trade improvement in fit against additional parameters.

1.9 What Does k Count?

The quantity k denotes the number of free parameters estimated in the likelihood. For the degree-p polynomial model

$$\mu_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p,$$

there are

$$p + 1$$

regression coefficients: $\beta_0, \beta_1, \ldots, \beta_p$. If the Gaussian variance σ^2 is also unknown, it is an additional estimated parameter. Therefore,

$$k = (p + 1) + 1 = p + 2.$$

More generally,

$$k = \text{total number of free parameters estimated by maximum likelihood.}$$

1.10 Connection AIC/BIC to Cross-Validation

Cross-validation and information criteria address the same general problem:

$$\boxed{\text{How complex should the fitted model be?}}$$

but they approach it differently. Cross-validation estimates predictive performance by repeatedly evaluating the fitted model on observations/data not used for fitting. AIC and BIC instead use the maximized likelihood on the observed data together with an explicit penalty for model complexity. Schematically,

$$\begin{aligned} \text{Cross-validation} &: \text{evaluate prediction on held-out data,} \\ \text{AIC/BIC} &: \text{training likelihood} + \text{explicit complexity penalty.} \end{aligned}$$